

ANALYSIS OF THREADED JOINTS IN KOLSKY BAR EXPERIMENTS

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ABSTRACT. // DRAFT - WORK-IN-PROGRESS - NOT FOR RELEASE // As of this writing, the intended audience for this brief is limited to the author.

1. INTRODUCTION

2. BACKGROUND

Vigil *et al.*¹ modeled a threaded screw as a three-dimensional wedge domain composed of a screw and substrate (see Figure 2). The Allow A-286 screw had material properties² $E = 29\text{e}+6$ psi, $\nu = 0.28$, $\sigma_Y = 120\text{e}+3$ psi, and $\sigma_U = 160\text{e}+3$ psi. The 304L stainless steel substrate had material properties $E = 28\text{e}+6$ psi, $\nu = 0.28$, $\sigma_Y = 32.7\text{e}+3$ psi, and $\sigma_U = 78\text{e}+3$ psi.

Homogeneous Dirichlet boundary conditions were applied to the longitudinal faces as required to allow the substrate to remain stationary while the screw to move axially. An inhomogeneous Dirichlet boundary condition was imposed on the top face screw, from zero to 0.635 mm (0.025 inches), shown (at maximum displacement) in Figure 2.

Castelluccio *et al.*³ examined model parameters of geometry (shape, size, tolerance), materials (anisotropy, inhomogeneity), mechanics (interface friction, temperature, loading) and methodology (discretization errors, coding implementation errors) to analyze sensitivity, error, and uncertainty on the force-displacement response of threaded fasteners.

They considered the #0-40 UNF bolt (see Figure 2) for the simulations.

Castelluccio performed a systematic comparison of 1D, 2.5D and 3D finite element models (see Figure 2).

¹Vigil MS, Brake MRW, VanGoethem DJ (2015) *Analysis of Modeling Parameters of Threaded Screws*, Sandia Report SAND2015-5135, June.

²Young's modulus, Poisson's ratio, yield stress, and ultimate tensile strength (respectively).

³Castelluccio GM, Brake MRW (2017 in press) *On the Origin of Computational Model Sensitivity, Error, and Uncertainty in Threaded Fasteners*, submitted to Computers & Structures, Second Preprint December 9, 2016.

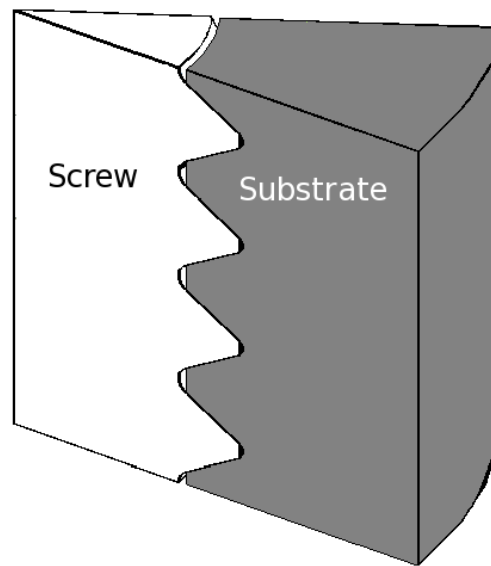


FIGURE 1. Reproduction of Vigil, Figure 3.

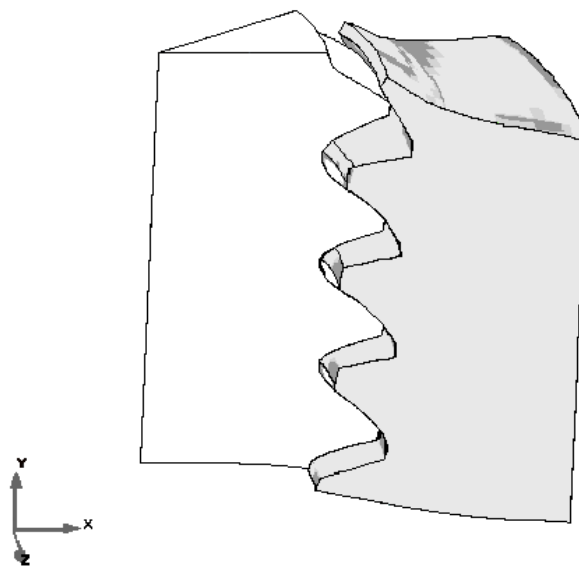


FIGURE 2. Reproduction of Vigil, Figure 5.

3. METHODS

“The incident bar was C300 maraging steel and 6061-T6 Aluminum I believe.” (see Sanborn 2017-01-25)

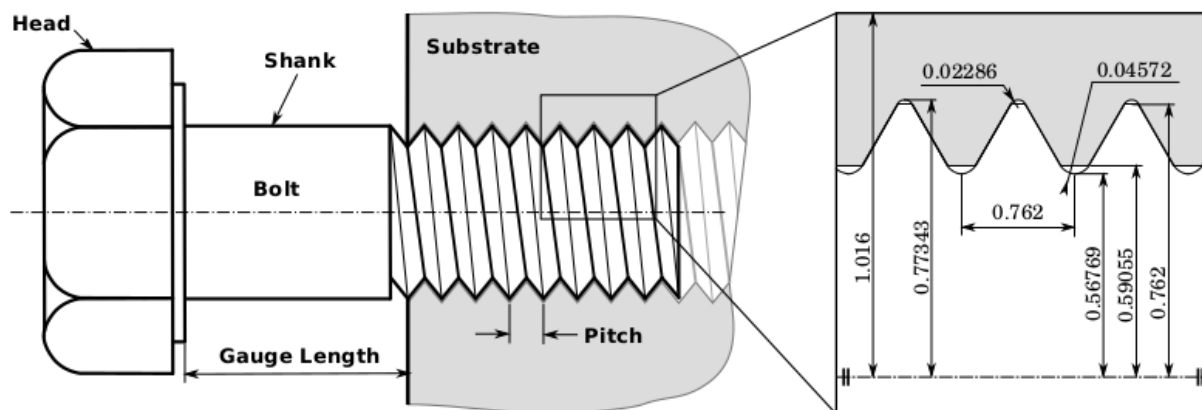
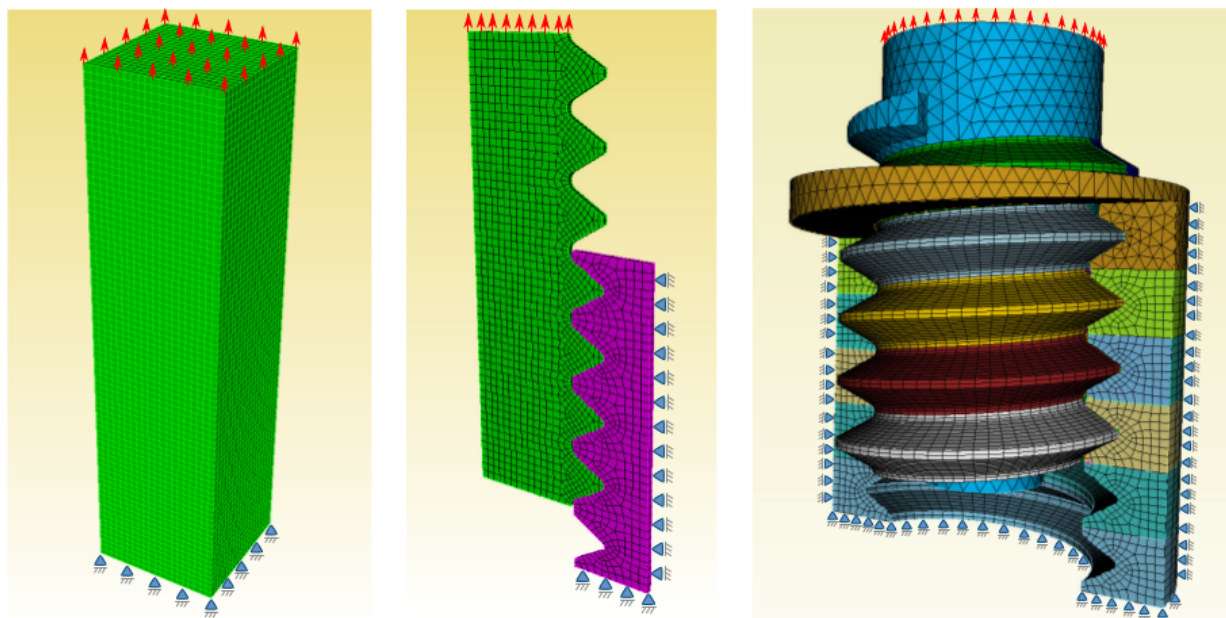


FIGURE 3. Reproduction of Castelluccio, Figure 2, showing the anatomy and geometry of the #40 UNF bolt (IS units).



(a) 1D model.

(b) 2.5D model.

(c) Fully 3D model.

FIGURE 4. Reproduction of Castelluccio, Figure 4, showing the 1D, 2.5D, and 3D models.

4. RESULTS

5. DISCUSSION

6. CONCLUSION

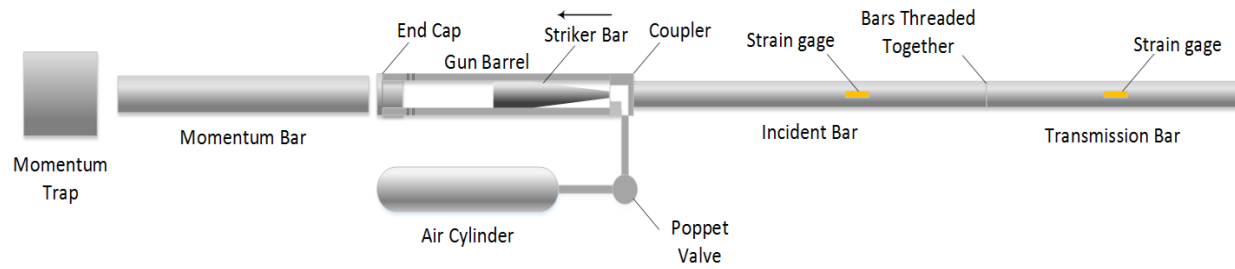


FIGURE 5. Hopkinson bar schematic.

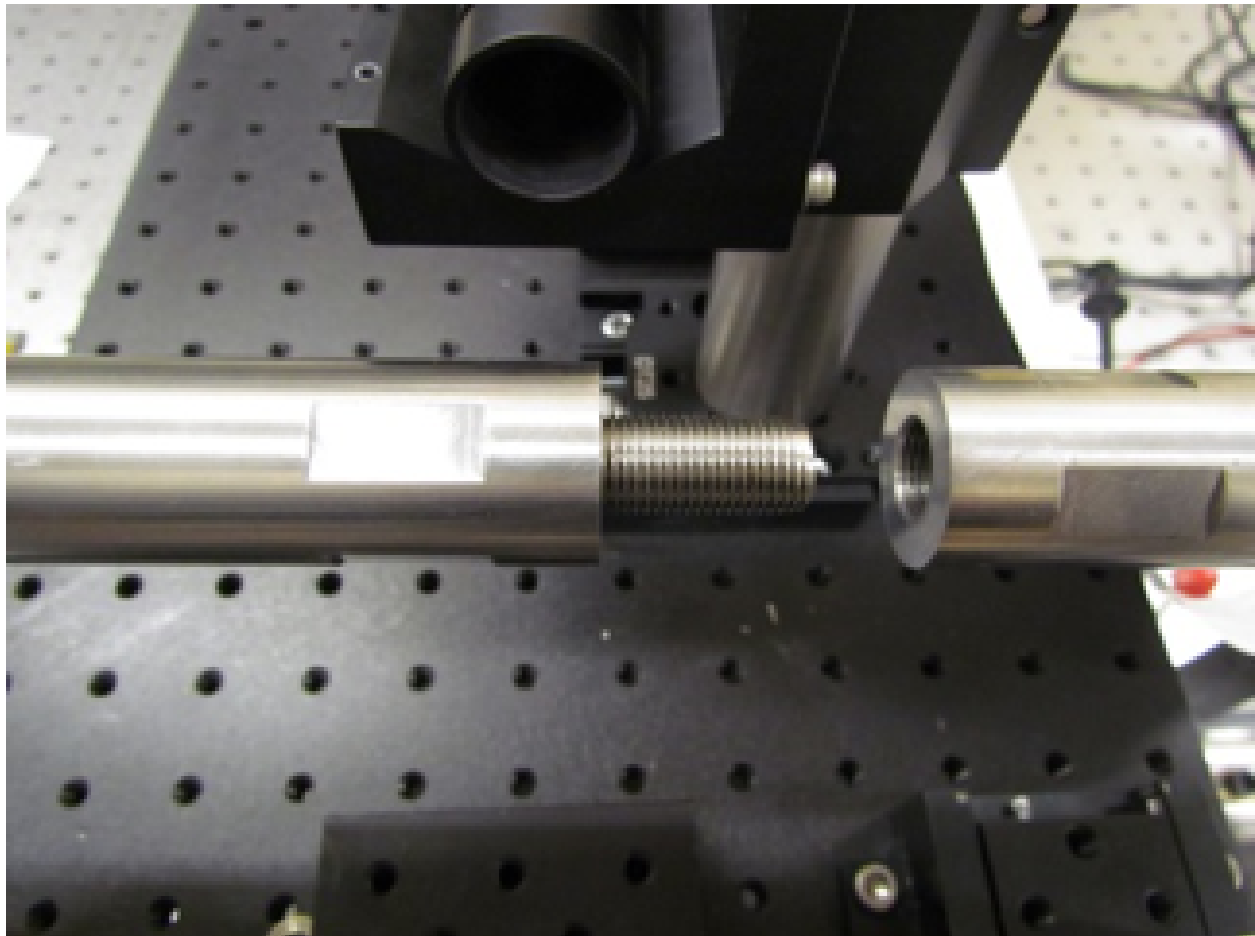


FIGURE 6. Close up view of threaded joint, prior to insertion.

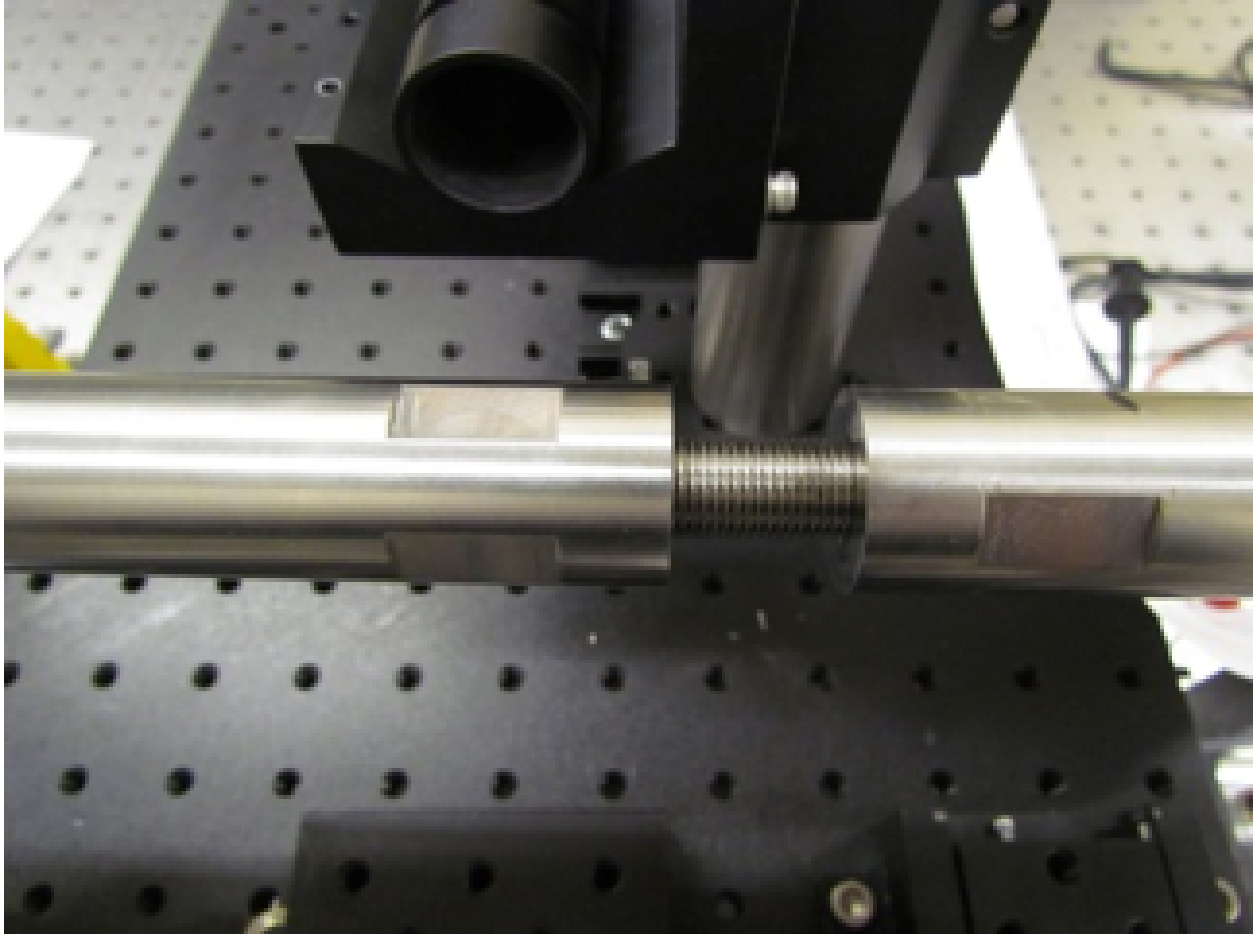


FIGURE 7. Close up view of threaded joint, during insertion.

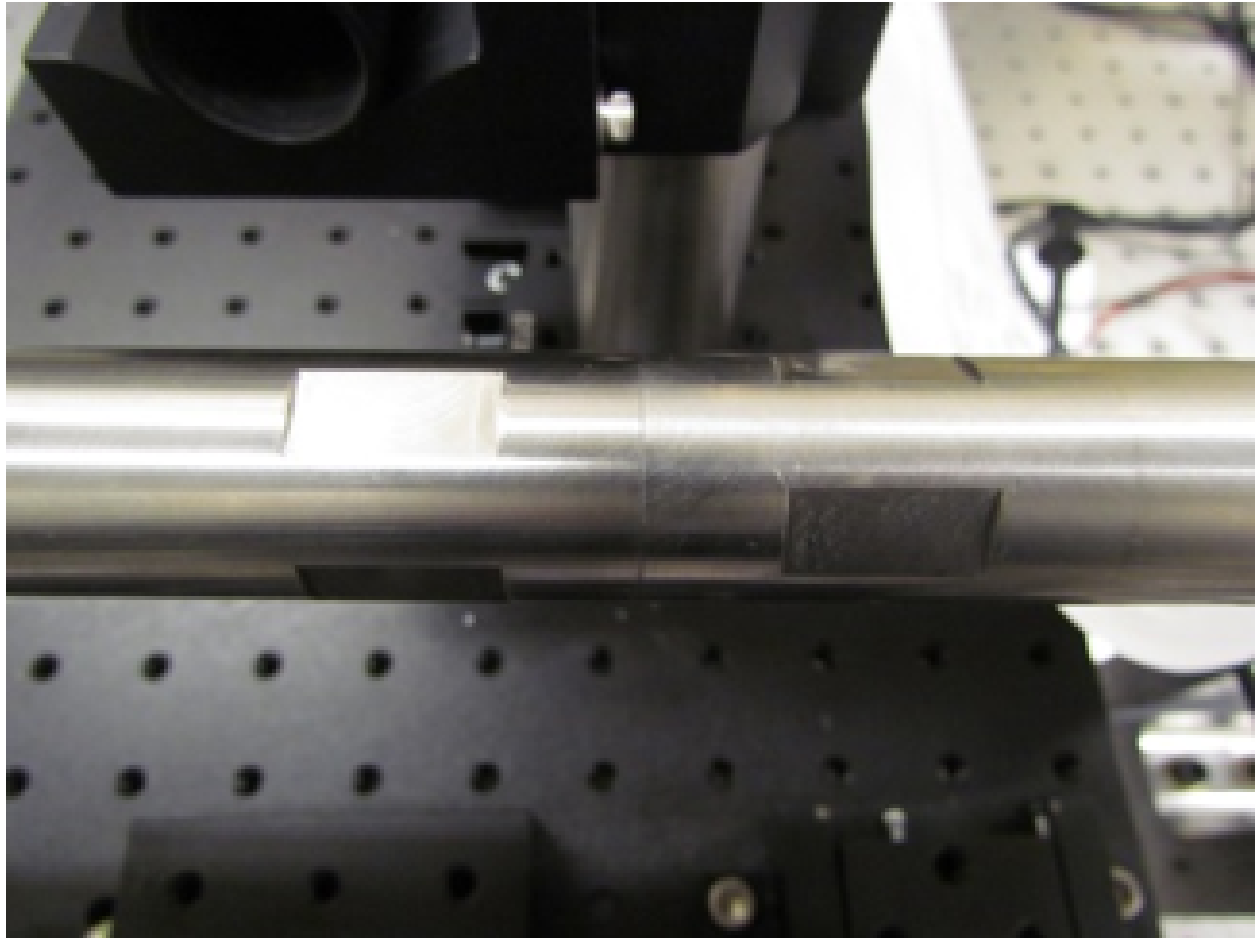


FIGURE 8. Close up view of threaded joint, after insertion.



FIGURE 9. Schematic showing incident bar as aluminum and transmission bar as steel.

7. REFERENCES

- Sanborn 2016-11-09. Sanborn B, Song B. (2016) Energy Dissipation through Threaded Joint Interfaces under Impact Tensile Loading, Fall 2016 TCG-XI Joint Meeting, November 9, 2016, Albuquerque, NM. (“TCG-XI Fall 2016 threaded joint.pptx”)
- Sanborn 2017-01-25. Sanborn B. E-mail communication.

APPENDIX A. COMMUNICATIONS

On 19 January 2017, I emailed Garth Reese (gmreese@sandia.gov), Matthew Brake (brake@rice.edu), and Douglas VanGoethem (djvango@sandia.gov), to inquire if they knew of additional modeling work in Cubit or simulation work in Sierra.

- REESE, Garth email 2016-01-19 1405: “I have worked with Matt Brake in the past, but that has focused on reduced order models such as the Iwan model. I’m not an adagio expert. I don’t even recall reviewing this paper — which is kind of odd. I wonder if the authors got it mixed up? You might contact Nathan Crane.”
- BRAKE, Matthew email 2016-01-19 1439:
 - cc: Castelluccio, Gustavo M (gmcaste@sandia.gov); Smith, Jeffrey (jasmith@sandia.gov); Robertson, Brett Anthony (barober@sandia.gov)
 - “There are several efforts that have been combined this FY by my understanding. My former postdoc, Gustavo Castelluccio, is now working with Jeff Smith (in 1550) on the sequel to this and other projects. Gustavo recently finished a paper on our efforts for modeling a fastener in SIERRA, and the conclusions are that the material model is more important than the geometric model. i.e. if you were to use a model of a square peg instead of the fastener, but with the right material model (i.e. elastic plastic), that will give you a surprisingly close answer to the truth data. If, on the other hand, you use a very detailed geometry with a poor material model (i.e. elastic for something that will experience yielding), then you will have a very poor answer compared to the truth data. The paper is either in the third round of reviews or it’s been accepted for publication, I’m not sure which.
 - We’re also working on spinning up a few complimentary programs here at Rice, but I’ve not yet gotten a computer setup with SIERRA installed on it.
 - Another branch of research that my students and I worked on while I was still at Sandia was looking at developing a surrogate model for the fastener so that you could have a high fidelity characterization of the fastener’s response built into a single degree of freedom constitutive model that was directly inserted into an FEA model (Brett Robertson did that work for Jeff Smith and me). That showed some good agreement with modeling experiments, and would be an area that I’d hope to find funding to continue one day.
 - I hope that helps. Let me know if you’ve any other questions.”